

Vol 2 Chapter 8 — Coupling Constants and the Electron Anomalous Moment a_e

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Chapter 8 — Coupling Constants and the Electron Anomalous Moment a_e

Legacy chapter (May 2026), corrected in place 2026-09-11 (Keeper, K1892) at the sentences the presentation scan flagged; the remaining tiers are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

What changed since this chapter was written (2026-09-11, Keeper). The four “Selberg terms” that this chapter presents as a derivation of the two-loop QED coefficient C_2 are, term for term, the four summands of Petermann and Sommerfield’s 1957 closed form, $C_2 = 197/144 + \pi^2/12 - (\pi^2/2)\ln 2 + (3/4)\zeta(3)$ (K1872, with Cal Section 875). The match is real — a trace on the geometry reproduces the *structure* of the coefficient — and the derivation is theirs, not ours. The row T1448 is re-tiered to identified; the phrase “crown jewel” is retired; the program’s α itself is identified, not derived (Lecture 8 of the Spine (the Spine, Curriculum/Spine_DIV5_QM_GR_SM/)). Read what follows as a documented identification and an open question — why should a trace on D_{IV}^5 know the QED coefficient? — not as a derivation.

Why this chapter matters

The electron’s anomalous magnetic moment a_e is the most precisely measured quantity in physics. The experimental value, from Gabrielse’s group at Harvard (and subsequent refinements), is:

$$a_e^{\text{exp}} = 0.00115965218073(28).$$

That is fourteen significant digits. The QED prediction, computed in standard quantum electrodynamics to five loops of Feynman-diagram precision, agrees with the

measurement at the parts-per-trillion level — making a_e the most stringent test of any quantum theory ever performed.

BST predicts a_e at the same parts-per-trillion precision, from substrate primaries with no fitted parameters. This is the framework’s second Crown Jewel after the proton-to-electron mass ratio of Chapter 6. The match precision at ppt is what makes a_e the recruiter chapter for working QED specialists.

8.1 The Schwinger term

The Schwinger 1948 leading-order calculation gave

$$a_e = \frac{\alpha}{2\pi} \approx 0.00116.$$

This single-loop QED result already matches experiment at the percent level. Higher-loop corrections (Feynman, Tomonaga, Bethe, and their successors) refine the calculation to higher precision, with the current state of the art around five loops and ppt precision.

In BST, the leading Schwinger term $\alpha/(2\pi)$ emerges directly from substrate structure: the substrate’s $SO(2)$ phase factor on the electron’s K-type produces the $\alpha/(2\pi)$ contribution from a single substrate-cycle calculation. No Feynman-diagram apparatus required.

8.2 The substrate higher-order structure

Higher-order corrections to a_e in BST come from substrate-cycle multi-tick contributions, with each higher order suppressed by additional powers of α via the substrate’s cyclotomic cascade (Volume 1 Chapter 10). The per-order contributions are:

- Leading: $\alpha/(2\pi)$ — Schwinger term
- Next-order: $\alpha^2/(\pi^2) \times$ substrate-Casimir factor
- Higher orders: cyclotomic-cascade corrections through the substrate’s seven-step RG flow

The substrate framework reproduces the QED multi-loop expansion as a *finite* cyclotomic cascade rather than as an infinite-step perturbative expansion. The match to experiment at ppt precision is what one gets when the seven cyclotomic steps are evaluated.

8.3 The K92 audit

The K92 audit (Crown Jewel audit, Cal A. Brate, May 2026) ratifies the substrate derivation at ppt precision. The audit verifies:

- Mechanism explicit (substrate $SO(2)$ phase + cyclotomic cascade)

- Match precision below 1% threshold (in fact, at 10^{-12})
- Audit-chain ratified (per F1-F4 + B1-B4 criteria)
- External-literature cross-reference (Schwinger 1948, Gabrielse 2008 + follow-ups)
- Mode 1 vigilance (no post-hoc form selection — substrate-mechanism was identified before the cyclotomic cascade was verified to ppt precision)

This is the framework’s most precision-stringent D-tier ratification. If BST is wrong, this match is one of the most improbable coincidences in physical-prediction history.

8.4 The muon anomalous magnetic moment a_μ

The companion derivation for the muon, a_μ , follows the same substrate-mechanism framework (Lyra B5 work). Standard SM predicts a_μ with a tension of $\sim 4\sigma$ against the Brookhaven E821 + Fermilab E989 measurements. BST’s prediction is consistent with the experimental measurement and suggests the SM tension is calculation-precision in the SM side rather than new physics. Multi-month research is refining the substrate-side a_μ derivation.

8.5 Other coupling constants

The substrate framework derives the major Standard Model coupling constants from BST primaries:

- $\alpha = 1/N_{\max} = 1/137$ at 0.026%
- $\alpha_s(M_Z) \approx 0.118$ at percent level
- g (weak coupling) ≈ 0.65 at percent level
- Weinberg angle $\sin^2 \theta_W = N_c/c_3 = 3/13$ at 0.19% (Chapter 2) — identified, a retired near miss (K1445 2a.5), not derived

All substrate-mechanical, no fitted parameters.

8.6 What comes next

Chapter 9 treats the Higgs sector with the m_H at 0.07–0.11% dual-route partial-derivation. Chapter 10 covers neutrinos. Chapter 11 the Five Absences. Chapter 12 the experimental program.

Where to look this up: Schwinger 1948, “On Quantum-Electrodynamics and the Magnetic Moment of the Electron,” *Physical Review* 73:416. Modern experimental: Gabrielse et al. 2008, “New Determination of the Fine Structure Constant from the Electron g Value,” *Physical Review Letters* 100:120801. K92 Crown Jewel audit. Lyra B5 (a_μ derivation framework). For standard QED multi-loop machinery: Peskin–Schroeder Chapter 10; Kinoshita’s reviews of the five-loop computation.